At most once includes never.

A lost tool call can hide zero accepted effects or one accepted effect.

EXECUTION CEILING

At most once

- Same deployed call is not replayed
- Limits one kind of duplication
- Does not promise an effect

KNOWLEDGE GAP

Zero or one

- Effect may not exist
- Effect may exist without its reply
- Destination evidence decides

Managed Agents and the Agents SDK were in public preview.

> **A retry count is not a destination receipt.**

The same lost call. Opposite realities.

From inside the session, both can look identical after a crash.



Conceptual state split; no live side-effect experiment was run.

The destination holds the missing evidence.

Three records answer three different questions.

Durable agent state and safe external action meet only when their identities join.



A stable operation ID is useful only if the receiving service recognizes it.

Keep progress, intent and acceptance separate.

Four statements.
Four different guarantees.

Reliable recovery begins by naming what each piece of evidence can and cannot establish.

EVIDENCE	ANSWERS	DOES NOT PROVE
Finished checkpoint	Where to resume	Outside effect accepted
At most once	Call did not run twice	Call ran once
Operation key	Same intended operation	Service accepted it
Destination receipt	Observed acceptance state	All later work completed

Unknown remains unknown when the destination cannot answer.

> **Do not promote a call property into an effect claim.**

Retry the operation, not a fresh intention.

A bounded recovery path asks the receiving system before creating another effect.

01

1 · Identify

Create one stable operation ID before the call.

02

2 · Send

Pass that identity to the receiving service.

03

3 · Read back

Query the destination after a lost response.

04

4 · Decide

Continue if accepted; retry same ID if absent.

05

5 · Hold

Retain UNKNOWN when the service cannot answer.

Without idempotency support, check first.

> **Same identity first; retry only after destination evidence.**

The bookmark worked. The effect test did not run.

The source microtest isolated durable local progress by using a read-only tool.

OBSERVED

Local read-only resume

One SHA-256 tool execution

Zero repeated tool actions

Zero external side effects

STILL UNKNOWN

Live effect recovery

Hosted-runtime crash behavior

Service deduplication and readback

Cost and completion under failure

No provider account, model call, payment, message, transaction or external action was used.

Narrow evidence is stronger when its boundary stays visible.

The retry count is not the receipt.

At most once limits repetition. The destination tells you whether once happened.

WEAK QUESTION

How many times did the agent try?

What did the destination accept under this operation ID?

After a lost call, read the receiving system before declaring zero, one or complete.

Local read-only test · External effect recovery UNKNOWN

> **Bind recovery to one operation identity and one observable acceptance record.**